

Bass Model Homework: Claude Sonnet 4

1. Chosen Innovation

The innovation selected for this analysis is **Claude Sonnet 4**, a general-purpose large language model developed by Anthropic and featured in TIME's Best Inventions of 2025:

<https://time.com/collections/best-inventions-2025/7318244/anthropic-claude-sonnet-4/>

Claude Sonnet 4 was released in May 2025 and saw rapid adoption: according to Anthropic, 75% of its API users switched to the new model within a week of release. It became especially popular among developers, and after release its context window was expanded so it could process much longer inputs, such as large codebases. Anthropic followed it with Claude Sonnet 4.5 in September 2025 [1].

2. Look-alike Innovation: The Internet

The Internet is selected as the look-alike innovation for Claude Sonnet 4. Both are general-purpose information technologies rather than single-function products: people use them to search for information, communicate, write, learn, and work. Like the Internet, large language models are accessed through devices that users already own, such as computers and smartphones, so adoption is limited mainly by awareness, perceived usefulness, and learning, not by the cost of new hardware.

The two innovations also share the same diffusion mechanism. In both cases a small group of technical early adopters is followed by mass adoption driven by word of mouth and social influence, which corresponds directly to the coefficients of innovation (p) and imitation (q) in the Bass model [2]. Finally, both target the general population worldwide, and adoption can be measured in the same way: as the share of individuals using the technology. Internet adoption has been recorded annually by the International Telecommunication Union (ITU) for more than three decades [3], providing a complete S-shaped diffusion curve on which the Bass model parameters can be estimated.

ChatGPT would be an even closer look-alike, but it was released only in late 2022, so its adoption history is too short to estimate the Bass parameters. One difference is that the Internet required new infrastructure, while Claude runs on devices people already own, so Claude may spread faster than the Internet did.

3. Historical Data

The data for the look-alike innovation is the annual share of the population using the Internet, collected by the International Telecommunication Union (ITU) and published by Our World in Data [3]. The dataset covers individual countries, regions, and the world as a whole from 1990 onward. The analysis below uses the US series; the choice of scope is discussed in Sections 3.1 and 6.

Source file: `data/share-of-individuals-using-the-internet.csv`

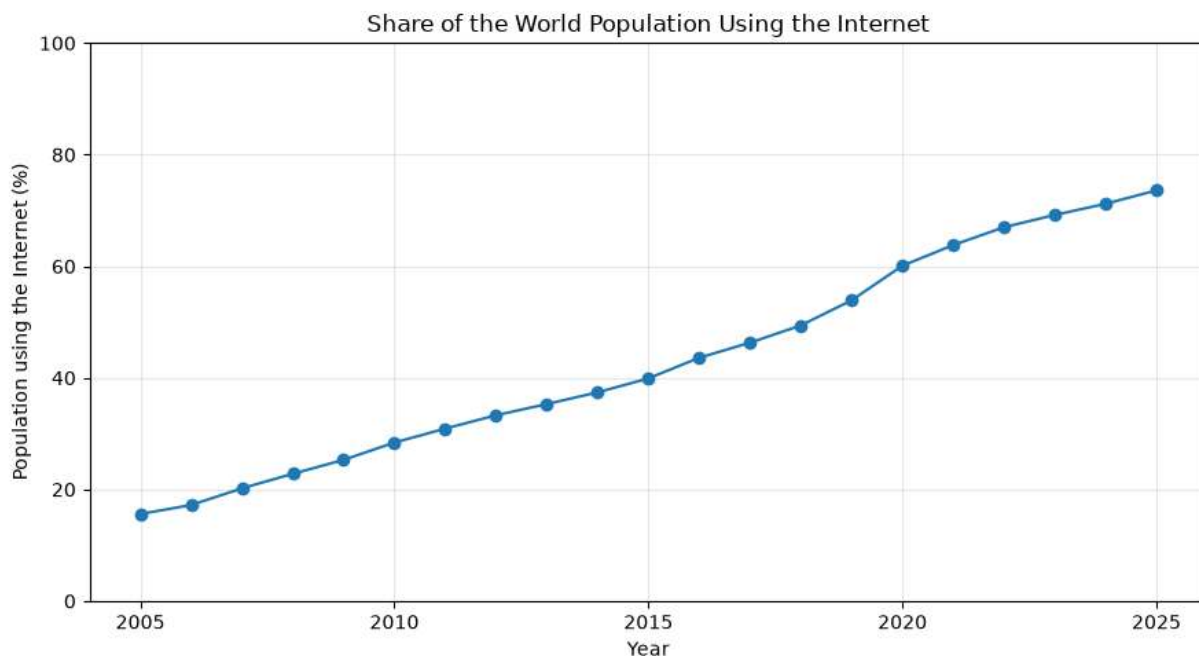
```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("../data/share-of-individuals-using-the-internet.csv")
df.columns = ["entity", "code", "year", "share_pct"]

world = df[df["entity"] == "World"][["year", "share_pct"]].reset_index(drop=True)
display(world)
```

	year	share_pct
0	2005	15.6
1	2006	17.2
2	2007	20.2
3	2008	22.8
4	2009	25.3
5	2010	28.4
6	2011	30.9
7	2012	33.3
8	2013	35.3
9	2014	37.4
10	2015	39.9
11	2016	43.6
12	2017	46.3
13	2018	49.4
14	2019	53.9
15	2020	60.1
16	2021	63.8
17	2022	67.0
18	2023	69.2
19	2024	71.2
20	2025	73.6

```
In [2]: fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(world["year"], world["share_pct"], marker="o", color="tab:blue")
ax.set_title("Share of the World Population Using the Internet")
ax.set_xlabel("Year")
ax.set_ylabel("Population using the Internet (%)")
ax.set_ylim(0, 100)
ax.set_xticks(range(2005, world["year"].max() + 1, 5))
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("../img/internet_adoption_world.png", dpi=150)
plt.show()
```



Let's see the historical dataset description.

```
In [3]: # Overview of the full dataset
print("Full dataset")
print(f"  Rows: {len(df)}")
print(f"  Entities (countries and regions): {df['entity'].nunique()}")
print(f"  Years covered: {df['year'].min()}-{df['year'].max()}")
print(f"  Missing values: {df['share_pct'].isna().sum()}")

# Overview of the worldwide series used in the analysis
expected_years = set(range(world["year"].min(), world["year"].max() + 1))
missing_years = sorted(expected_years - set(world["year"]))

print("\nWorldwide series")
print(f"  Years covered: {world['year'].min()}-{world['year'].max()} ({len(world)})")
print(f"  Missing years: {missing_years if missing_years else 'none'}")
print(f"  First value: {world['share_pct'].iloc[0]:.2f}% ({world['year'].iloc[0]})")
print(f"  Last value: {world['share_pct'].iloc[-1]:.2f}% ({world['year'].iloc[-1]})

# Yearly increase in adoption (percentage points)
world["yearly_increase_pp"] = world["share_pct"].diff()
peak = world.loc[world["yearly_increase_pp"].idxmax()]
print(f"  Largest yearly increase: {peak['yearly_increase_pp']:.2f} pp in {int(peak['year'])}")

display(world["share_pct"].describe().round(2).to_frame("share_pct (%)"))
```

Full dataset

Rows: 6659

Entities (countries and regions): 231

Years covered: 1990–2025

Missing values: 0

Worldwide series

Years covered: 2005–2025 (21 observations)

Missing years: none

First value: 15.60% (2005)

Last value: 73.60% (2025)

Largest yearly increase: 6.20 pp in 2020

share_pct (%)	
count	21.00
mean	43.07
std	18.84
min	15.60
25%	28.40
50%	39.90
75%	60.10
max	73.60

```
In [4]: for country in ["United States", "Armenia"]:
        s = df[df["entity"] == country][["year", "share_pct"]]
        expected = set(range(s["year"].min(), s["year"].max() + 1))
        print(f"{country}: {s['year'].min()}--{s['year'].max()}, "
              f"{len(s)} observations, missing years: {sorted(expected - set(s['year']))}")
        print(f"    {s['share_pct'].iloc[0]:.2f}% → {s['share_pct'].iloc[-1]:.2f}%\n")
```

United States: 1990–2024, 35 observations, missing years: []
 0.79% → 94.69%

Armenia: 1990–2024, 32 observations, missing years: [1991, 1992, 1993]
 0.00% → 81.34%

3.1 Data Overview

The dataset contains 6,659 annual observations for 231 countries and regions (1990–2025), with no missing values. The worldwide series covers 2005–2025: adoption grew from 15.6% to 73.6%, with the largest yearly increase (6.2 pp) in 2020, likely driven by the COVID-19 pandemic. However, this series starts when 15.6% of the world was already online, so it misses the early diffusion phase that the Bass model relies on to estimate p . The analysis therefore uses the United States series, which is complete from 1990 to 2024 (0.79% → 94.69%, no missing years).

4. Estimating the Bass Model Parameters

```
In [5]: import sys
sys.path.append("..")
from helper_functions import bass_F, bass_cumulative, fit_bass, r_squared, peak_time
import numpy as np
```

The Bass model [2] describes cumulative adoption as $A(t) = M \cdot F(t)$, where

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}$$

p is the coefficient of innovation, q is the coefficient of imitation, and M is the market potential. The parameters are estimated by non-linear least squares on the US cumulative adoption series, with $t = 1$ in 1990. M is expressed as a share of the population and is bounded by 100%.

```
In [6]: us = df[df["entity"] == "United States"][["year", "share_pct"]].reset_index(drop=True)
launch_year = us["year"].min()
t = us["year"] - launch_year + 1

p, q, m = fit_bass(t, us["share_pct"], m_upper=100)
us["fitted_pct"] = bass_cumulative(t, p, q, m)
r2 = r_squared(us["share_pct"], us["fitted_pct"])
t_peak = peak_time(p, q)

results = pd.DataFrame({
    "Parameter": ["p (innovation)", "q (imitation)",
                  "M (market potential, % of population)", "R²", "Peak adoption year"],
    "Value": [round(p, 4), round(q, 4), round(m, 2), round(r2, 4),
              round(launch_year - 1 + t_peak, 1)],
})
display(results)
```

	Parameter	Value
0	p (innovation)	0.0248
1	q (imitation)	0.1321
2	M (market potential, % of population)	94.6900
3	R²	0.9614
4	Peak adoption year	1999.7000

```
In [7]: fig, axes = plt.subplots(1, 2, figsize=(13, 4.5))

axes[0].plot(us["year"], us["share_pct"], "o", label="Actual")
axes[0].plot(us["year"], us["fitted_pct"], "-", label="Bass model fit")
axes[0].set_title("Cumulative Internet Adoption, US")
```

```

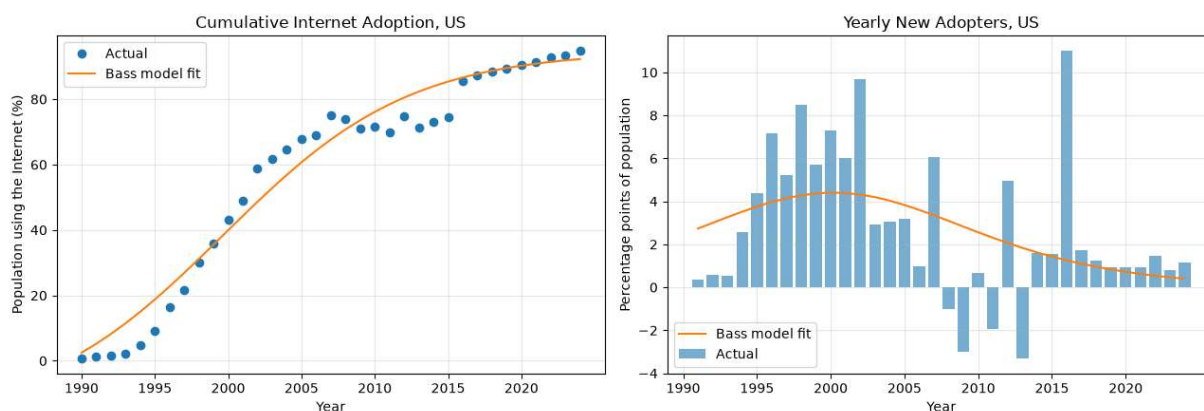
axes[0].set_ylabel("Population using the Internet (%)")

axes[1].bar(us["year"], us["share_pct"].diff(), alpha=0.6, label="Actual")
axes[1].plot(us["year"], us["fitted_pct"].diff(), "-", color="tab:orange", label="B")
axes[1].set_title("Yearly New Adopters, US")
axes[1].set_ylabel("Percentage points of population")

for ax in axes:
    ax.set_xlabel("Year")
    ax.set_xticks(range(1990, 2025, 5))
    ax.grid(alpha=0.3)
    ax.legend()

plt.tight_layout()
plt.savefig("../img/bass_fit_internet_us.png", dpi=150)
plt.show()

```



The model fits the data well ($R^2 = 0.96$): it captures the overall S-shape of Internet adoption in the US, with yearly adoption peaking around 2000. The coefficient of imitation is about five times larger than the coefficient of innovation, which means adoption was driven mainly by word of mouth and social influence rather than by independent early adopters. Compared with the average values reported in a meta-analysis of diffusion studies ($p = 0.03$, $q = 0.38$) [7], the Internet shows a similar innovation effect but a slower imitation effect, which spreads adoption over a longer period.

The estimated market potential M equals the latest observed value (94.69%). This is the lower bound set in the estimation, since market potential cannot be smaller than the number of people who have already adopted.

5. Predicting the Diffusion of Claude Sonnet 4

Assuming that Claude Sonnet 4 diffuses in the same way as its look-alike, the coefficients estimated for the Internet are applied to Claude Sonnet 4, with $t = 1$ in 2025, its year of release [1]. The forecast shows the share of the market potential M that adopts each year, and the cumulative share $F(t)$. The size of M for Claude Sonnet 4 is estimated in Sections 6–7.

```
In [8]: claude_launch = 2025
forecast_years = np.arange(claude_launch, claude_launch + 26)
t_f = forecast_years - claude_launch + 1

forecast = pd.DataFrame({
    "year": forecast_years,
    "new_adopters_pct_of_M": (bass_F(t_f, p, q) - bass_F(t_f - 1, p, q)) * 100,
    "cumulative_pct_of_M": bass_F(t_f, p, q) * 100,
})
display(forecast.round(2))

peak_year = claude_launch - 1 + t_peak
half_year = forecast.loc[forecast["cumulative_pct_of_M"] >= 50, "year"].iloc[0]
print(f"Predicted peak of yearly adoption: {peak_year:.1f}")
print(f"Half of the market potential reached by: {half_year}")
```

	year	new_adopters_pct_of_M	cumulative_pct_of_M
0	2025	2.61	2.61
1	2026	2.89	5.50
2	2027	3.17	8.67
3	2028	3.45	12.12
4	2029	3.72	15.83
5	2030	3.97	19.80
6	2031	4.19	23.99
7	2032	4.38	28.37
8	2033	4.52	32.89
9	2034	4.62	37.51
10	2035	4.65	42.16
11	2036	4.63	46.80
12	2037	4.56	51.36
13	2038	4.43	55.79
14	2039	4.26	60.05
15	2040	4.05	64.09
16	2041	3.80	67.90
17	2042	3.54	71.44
18	2043	3.26	74.70
19	2044	2.98	77.68
20	2045	2.70	80.39
21	2046	2.44	82.82
22	2047	2.18	85.00
23	2048	1.94	86.94
24	2049	1.71	88.65
25	2050	1.51	90.16

Predicted peak of yearly adoption: 2034.7

Half of the market potential reached by: 2037

```
In [9]: fig, ax1 = plt.subplots(figsize=(10, 5))
ax1.bar(forecast["year"], forecast["new_adopters_pct_of_M"], alpha=0.6, label="New
ax1.set_ylabel("New adopters (% of M)")

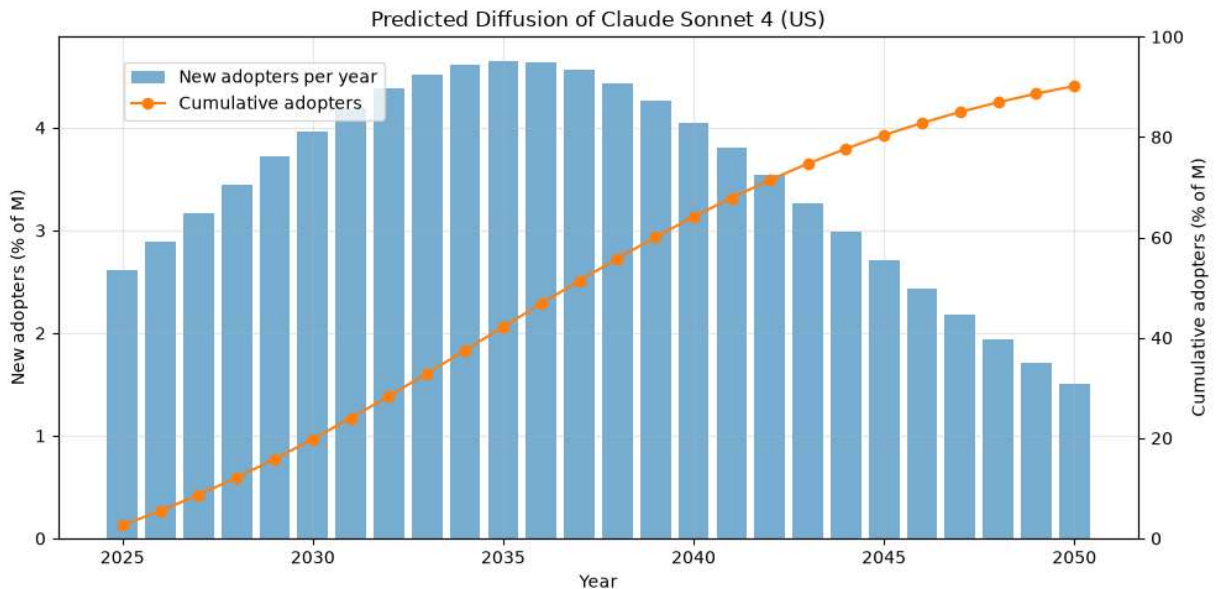
ax2 = ax1.twinx()
```

```

ax2.plot(forecast["year"], forecast["cumulative_pct_of_M"], "o-", color="tab:orange")
ax2.set_ylabel("Cumulative adopters (% of M)")
ax2.set_ylim(0, 100)

ax1.set_title("Predicted Diffusion of Claude Sonnet 4 (US)")
ax1.set_xlabel("Year")
ax1.set_xticks(forecast["year"][::5])
ax1.grid(alpha=0.3)
fig.legend(loc="upper left", bbox_to_anchor=(0.1, 0.9))
plt.tight_layout()
plt.savefig("../img/claude_diffusion_forecast.png", dpi=150)
plt.show()

```



5.1 Results

Although Claude Sonnet 4 is already on the market, its adoption history is too short to estimate the Bass parameters directly. The parameters of its look-alike, the Internet, are therefore used to forecast its diffusion.

As shown in the forecast table and figure above, the predicted adoption follows the typical S-shaped curve: 2.61% of the market potential adopts in 2025, and cumulative adoption reaches 19.80% of M by 2030. Yearly adoption peaks around 2035 (4.65% of M), after which it gradually declines as the market approaches saturation. The low coefficient of imitation ($q = 0.1321$) results in a long diffusion period, so this forecast can be seen as a conservative scenario.

Since Claude Sonnet 4 was followed by Claude Sonnet 4.5 within a few months [1], the forecast should be read as the adoption of Claude as a product, not of one specific model version. Users of Sonnet 4 move to newer versions, but they remain Claude adopters.

6. Scope: United States

The analysis focuses on the United States rather than the whole world, for three reasons.

First, the US is Claude's largest market. According to the Anthropic Economic Index, the United States accounts for 21.6% of global Claude.ai usage, the highest share of any country, ahead of India, Brazil, Japan and South Korea [4]. Anthropic is also a US-based company, so the US is its home and primary market.

Second, the data supports a country-level analysis. As shown in Section 3, the US Internet adoption series is complete for 1990–2024, while the worldwide series starts only in 2005 and misses the early diffusion phase.

Third, a single country gives a well-defined market. Its population is known, so the market potential can be translated into a number of adopters (Section 7), and the look-alike and the forecast refer to the same population.

7. Number of Adopters by Period

I found out that the market potential M is estimated using Fermi logic. The US population was 341,784,857 on July 1, 2025, of which 21.1% were under 18 [5]. Since 44% of US adults use ChatGPT [6], this share is used as a proxy for adults who use AI assistants and could therefore adopt Claude. The number of new adopters in year t is $a(t) = M \cdot (F(t) - F(t - 1))$.

```
In [10]: us_population = 341_784_857
share_adults = 1 - 0.211
share_ai_users = 0.44

M_claude = us_population * share_adults * share_ai_users
print(f"Market potential M: {M_claude / 1e6:.1f} million adults")

forecast["new_adopters_mln"] = M_claude * forecast["new_adopters_pct_of_M"] / 100 /
forecast["cumulative_adopters_mln"] = M_claude * forecast["cumulative_pct_of_M"] /
display(forecast[["year", "new_adopters_mln", "cumulative_adopters_mln"]].round(2))
```

Market potential M: 118.7 million adults

	year	new_adopters mln	cumulative_adopters mln
0	2025	3.10	3.10
1	2026	3.43	6.53
2	2027	3.76	10.29
3	2028	4.09	14.38
4	2029	4.41	18.79
5	2030	4.71	23.49
6	2031	4.97	28.47
7	2032	5.20	33.66
8	2033	5.37	39.03
9	2034	5.48	44.51
10	2035	5.52	50.03
11	2036	5.50	55.53
12	2037	5.41	60.94
13	2038	5.26	66.20
14	2039	5.05	71.25
15	2040	4.80	76.05
16	2041	4.51	80.56
17	2042	4.20	84.76
18	2043	3.87	88.64
19	2044	3.54	92.17
20	2045	3.21	95.38
21	2046	2.89	98.27
22	2047	2.59	100.86
23	2048	2.30	103.16
24	2049	2.03	105.19
25	2050	1.79	106.98

The estimated market potential is 118.7 million US adults. The model predicts that 3.10 million adults start using Claude in 2025, and about 23.5 million in total by 2030. The number of new users grows every year until 2035, when it peaks at 5.52 million, and by then about half of the market potential (50.03 million people) has adopted Claude. After 2035, yearly adoption slowly declines as fewer potential users remain.

Conclusion

Using the Internet as a look-alike innovation, the Bass model was estimated on US Internet adoption data ($p = 0.0248$, $q = 0.1321$, $R^2 = 0.96$) and applied to Claude. With a market potential of 118.7 million US adults, the forecast shows adoption growing steadily from 3.10 million new users in 2025 to a peak of 5.52 million in 2035. Since adoption is driven mainly by imitation ($q > p$), word of mouth and social influence are expected to be the main drivers of Claude's growth.

References

- [1] Hays, K. (2025, October 9). Anthropic Claude Sonnet 4. *TIME Best Inventions 2025*. <https://time.com/collections/best-inventions-2025/7318244/anthropic-claude-sonnet-4/>
- [2] Bass, F. M. (1969). A new product growth for model consumer durables. *Management Science*, 15(5), 215–227.
- [3] International Telecommunication Union (ITU), via Our World in Data. (2025). Share of the population using the Internet [Data set]. <https://ourworldindata.org/grapher/share-of-individuals-using-the-internet>
- [4] Anthropic. (2025, September 15). Anthropic Economic Index: Tracking AI's role in the US and global economy. <https://www.anthropic.com/research/economic-index-geography>
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- [7] Sultan, F., Farley, J. U., & Lehmann, D. R. (1990). A meta-analysis of applications of diffusion models. *Journal of Marketing Research*, 27(1), 70–77.